TWO OUT OF THREE FOR RIGHT ACYCLIC OBJECTS IN A SHORT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5880-5895, following the statement at 5860-5878.

Editorial note. No omissions: the triangle and LES are assembled from named lemmas and the three cases are read off the LES term by term.

Allowed standard prerequisites (editorial).

- Canonical triangle of a termwise split sequence. lemma-derived-canonical-delta-functor (carded L26230, cited at ~5882). Produces the triangle (A, B, C, a, b, c).
- Cohomology of the derived functor. lemma-left-exact-higher-derived (carded L26184, cited at ~5886). R^iF = 0 for i < 0 and R^0F = F; RF exact gives the LES.
- Characterization of right acyclic objects. lemma-F-acyclic (same chapter, cited at ~5894). Converts vanishing of higher derived functors into acyclicity.

Source reference. Original lemma 015C: sources/stacks-derived/pinned-original-source.tex, source lines 5836–5836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0152: sources/stacks-derived/pinned-original-source.tex, source lines 4182–4182. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TD: sources/stacks-derived/pinned-original-source.tex, source lines 5782–5782. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor between abelian categories and assume $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is everywhere defined. Let $0 \to A \to B \to C \to 0$ be a short exact sequence of $\mathcal{A}$.*

(1) *If A and C are right acyclic for F then so is B.*
(2) *If A and B are right acyclic for F then so is C.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) *If B and C are right acyclic for F and $F(B) \to F(C)$ is surjective then A is right acyclic for F.*

In each of the three cases

$$0 \to F(A) \to F(B) \to F(C) \to 0$$

is a short exact sequence of $\mathcal{B}$.

Proof. By Lemma 0152 we obtain a distinguished triangle $(A[0], B[0], C[0], a, b, c)$ in $D^+(\mathcal{A})$. As RF is an exact functor and since $R^iF = 0$ for $i < 0$ and $R^0F = F$ (Lemma 05TD) we obtain an exact cohomology sequence

$$0 \to F(A) \to F(B) \to F(C) \to R^1F(A) \to \ldots$$

in the abelian category $\mathcal{B}$. Thus the lemma follows from the characterization of acyclic objects in Lemma 015C. □